



IOTRICITY

S3

Team Name

Centurion

Problem Statement

Water tanks are often monitored manually, which can lead to overflow, low-water conditions, unnecessary pump operation, and wastage of water and electricity. Our project proposes an Arduino-based monitoring system that continuously checks the water condition and provides real-time status indication and automatic pump control.

Problem identification & analysis

The Problem

- i) Water tanks are commonly monitored manually.
- ii) Users may not know the current water level without physically checking the tank.
- iii) Low-water conditions can cause unnecessary pump operation or inconvenience.
- iv) Continuous monitoring of tank status is difficult without an automated system.

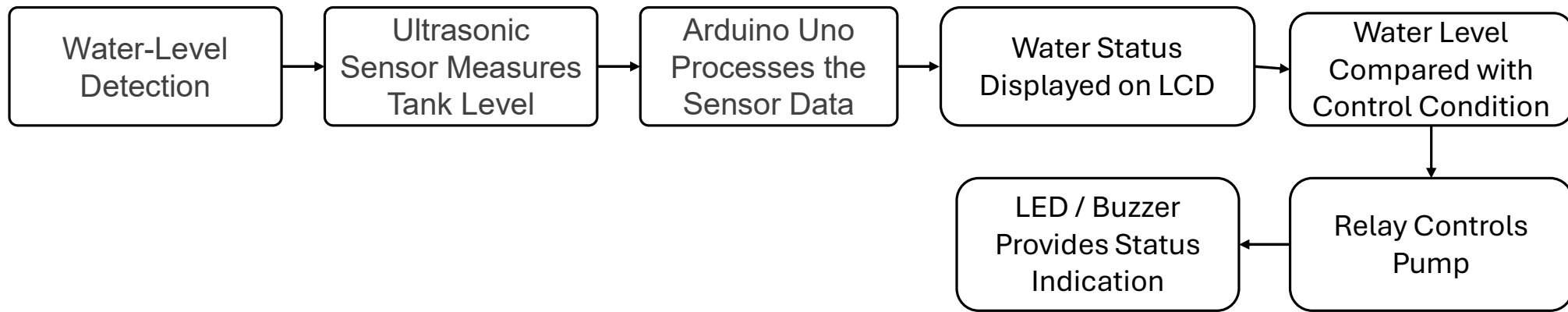
State the problem

“There is a need for an automated system that can monitor water-tank conditions and provide timely status information while controlling the pump according to the detected water level.”

Our Solution

We propose to develop a Smart Water Monitoring and Tank Safety System that uses sensors, an Arduino controller and output indicators to monitor water conditions and control the pump automatically.”

Proposed Technical Approach



Technology Stack

Hardware:

Arduino Uno :-Main controller that processes sensor data and controls outputs

Ultrasonic Sensor :-Detects the water level

16×2 LCD Display :-Displays water status and system information

Relay Module :-Provides switching/control for the pump

Buzzer:-Provides audible alerts

LED Indicators:-Shows different system/status conditions

Resistors & Jumper:-Wires Used for connections and LED current limiting.

Softwares:

- Arduino Programming / Embedded C/C++ — System control logic
- Wokwi — Circuit design and simulation
- Arduino simulation environment — Testing the system behaviour before physical implementation

Feasibility & Viability

Technical feasibility:

- The system can be developed using commonly available Arduino-compatible components.
- The required sensor, controller and output modules can be integrated using standard connections.
- The working concept has been demonstrated through Wokwi simulation.

Resources required:

- 1) Arduino Uno
- 2) Ultrasonic sensor
- 3) LCD display
- 4) Relay module
- 5) Buzzer
- 6) LEDs and resistors
- 7) Connecting wires
- 8) Water pump for physical implementation

Budget:

- 1) Low cost friendly budget
- 2) Estimated prices 1k to 2k

Time:

5 to 8 working days

Challenges:

- Accurate water-level measurement
- Reliable sensor readings
- Proper pump switching
- Preventing unwanted relay activation

Future scope:

- IoT/mobile monitoring
- Water-quality sensors
- Cloud data logging
- Multiple tank monitoring
- More advanced automatic pump control

Impact & Benefits

Target Beneficiaries:

- Households
- Hostels
- Schools and colleges
- Apartments
- Small commercial buildings

Key benefits:

- Automated water-level monitoring*
- Automatic pump control
- Real-time status display
- Visual and audible alerts
- Potential reduction in water and electricity wastage

Impact measurement:

The system can make water-tank management more automated, convenient and safety-oriented by continuously monitoring the tank and responding to low-water conditions.

Limitations & Future Scope:

- Add a turbidity sensor for water clarity monitoring.
- Add a TDS sensor for dissolved-solids monitoring.
- Add IoT connectivity for remote monitoring.
- Develop a mobile application for live sensor readings.